earlier_seen | equal-weighted | Absolute loss surfaces
Expanding controls are deduplicated sidebands, not finite coordinates
Exploratory seen-period estimates; calibration/power deferred; no calibrated significance claim

Separate | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.128	0.941	0.914	0.882	0.873	0.860	0.858	0.848	0.844	0.845	0.847	0.847	0.847	0.843	0.840	0.827	0.822	0.823	0.817
7	1.120	0.941	0.914	0.879	0.869	0.856	0.855	0.844	0.841	0.839	0.843	0.842	0.841	0.838	0.837	0.823	0.820	0.820	0.815
6	1.105	0.931	0.908	0.876	0.862	0.852	0.851	0.843	0.838	0.838	0.838	0.839	0.837	0.835	0.834	0.820	0.817	0.817	0.813
5	1.089	0.922	0.903	0.869	0.856	0.847	0.846	0.839	0.835	0.835	0.833	0.835	0.835	0.834	0.833	0.820	0.817	0.818	0.814
4	1.074	0.923	0.893	0.863	0.848	0.842	0.841	0.835	0.831	0.832	0.831	0.834	0.833	0.831	0.831	0.820	0.817	0.818	0.813
3	1.066	0.908	0.885	0.858	0.845	0.837	0.835	0.832	0.828	0.829	0.831	0.834	0.834	0.831	0.831	0.822	0.818	0.819	0.815
2	1.046	0.901	0.880	0.857	0.846	0.841	0.842	0.836	0.831	0.832	0.831	0.832	0.833	0.833	0.833	0.823	0.820	0.820	0.815
1	1.029	0.894	0.874	0.850	0.847	0.839	0.842	0.833	0.831	0.833	0.832	0.833	0.833	0.832	0.832	0.824	0.821	0.822	0.818

Coefficient-history pairs

Separate | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.227	0.990	0.934	0.903	0.884	0.869	0.864	0.854	0.849	0.846	0.847	0.850	0.850	0.847	0.845	0.830	0.825	0.825	0.818
7	1.197	0.978	0.927	0.896	0.877	0.864	0.860	0.850	0.846	0.844	0.846	0.849	0.848	0.845	0.842	0.828	0.823	0.823	0.817
6	1.166	0.959	0.916	0.883	0.866	0.854	0.853	0.845	0.840	0.839	0.840	0.843	0.844	0.842	0.840	0.826	0.821	0.821	0.815
5	1.123	0.944	0.909	0.874	0.860	0.849	0.848	0.842	0.839	0.838	0.838	0.841	0.840	0.839	0.838	0.826	0.822	0.822	0.815
4	1.101	0.928	0.897	0.865	0.853	0.844	0.845	0.839	0.837	0.838	0.837	0.840	0.839	0.838	0.837	0.826	0.822	0.823	0.817
3	1.088	0.924	0.895	0.862	0.853	0.843	0.847	0.837	0.834	0.834	0.835	0.836	0.836	0.835	0.835	0.825	0.821	0.822	0.816
2	1.066	0.918	0.890	0.862	0.856	0.846	0.846	0.835	0.834	0.835	0.834	0.836	0.835	0.834	0.834	0.825	0.823	0.823	0.818
1	1.025	0.893	0.873	0.849	0.846	0.838	0.841	0.832	0.830	0.832	0.831	0.832	0.832	0.832	0.832	0.823	0.820	0.821	0.817

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.808
7	0.808
6	0.807
5	0.808
4	0.812
3	0.812
2	0.814
1	0.813

Coefficient-history pairs

Common | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.036	0.899	0.875	0.852	0.844	0.838	0.840	0.830	0.826	0.829	0.832	0.833	0.835	0.833	0.832	0.823	0.820	0.821	0.815
7	1.034	0.897	0.875	0.848	0.842	0.838	0.837	0.828	0.825	0.828	0.832	0.833	0.832	0.832	0.832	0.823	0.820	0.820	0.815
6	1.028	0.894	0.871	0.845	0.839	0.835	0.834	0.827	0.825	0.828	0.829	0.832	0.831	0.831	0.831	0.820	0.818	0.818	0.813
5	1.022	0.889	0.868	0.841	0.834	0.830	0.832	0.825	0.824	0.828	0.827	0.830	0.831	0.831	0.830	0.820	0.819	0.820	0.815
4	1.014	0.891	0.865	0.841	0.831	0.829	0.830	0.824	0.823	0.826	0.826	0.830	0.830	0.829	0.830	0.820	0.817	0.819	0.813
3	1.012	0.883	0.862	0.837	0.831	0.828	0.827	0.823	0.821	0.824	0.827	0.830	0.830	0.829	0.830	0.822	0.817	0.819	0.814
2	1.013	0.883	0.863	0.842	0.834	0.833	0.837	0.829	0.826	0.827	0.828	0.830	0.831	0.831	0.832	0.823	0.820	0.821	0.817
1	1.006	0.884	0.862	0.840	0.839	0.833	0.838	0.829	0.827	0.830	0.829	0.831	0.832	0.832	0.832	0.825	0.822	0.823	0.819

Coefficient-history pairs

Common | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.068	0.913	0.879	0.857	0.846	0.840	0.838	0.831	0.828	0.829	0.831	0.835	0.836	0.835	0.834	0.824	0.820	0.821	0.815
7	1.052	0.907	0.876	0.852	0.842	0.837	0.837	0.828	0.826	0.828	0.831	0.835	0.835	0.834	0.833	0.823	0.819	0.820	0.814
6	1.052	0.904	0.873	0.850	0.839	0.835	0.836	0.828	0.825	0.828	0.829	0.833	0.834	0.834	0.833	0.823	0.819	0.821	0.814
5	1.036	0.901	0.874	0.849	0.840	0.835	0.835	0.828	0.826	0.828	0.830	0.834	0.834	0.834	0.833	0.823	0.820	0.821	0.815
4	1.026	0.893	0.867	0.843	0.835	0.830	0.832	0.825	0.824	0.828	0.829	0.833	0.833	0.833	0.832	0.823	0.819	0.821	0.815
3	1.027	0.894	0.869	0.843	0.838	0.833	0.838	0.829	0.826	0.828	0.829	0.832	0.832	0.833	0.833	0.824	0.821	0.822	0.817
2	1.029	0.898	0.872	0.847	0.843	0.838	0.841	0.830	0.829	0.831	0.831	0.833	0.833	0.834	0.834	0.826	0.824	0.825	0.821
1	1.002	0.882	0.861	0.839	0.838	0.832	0.836	0.828	0.827	0.829	0.829	0.831	0.831	0.832	0.832	0.824	0.822	0.823	0.819

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.809
7	0.809
6	0.809
5	0.809
4	0.812
3	0.813
2	0.815
1	0.815

Coefficient-history pairs

Partial | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.059	0.914	0.888	0.861	0.852	0.846	0.842	0.832	0.831	0.833	0.835	0.836	0.836	0.833	0.831	0.821	0.818	0.820	0.814
7	1.055	0.911	0.890	0.858	0.851	0.845	0.842	0.830	0.829	0.831	0.833	0.833	0.832	0.831	0.831	0.820	0.817	0.818	0.813
6	1.051	0.906	0.885	0.854	0.845	0.841	0.837	0.829	0.829	0.831	0.831	0.833	0.831	0.831	0.830	0.817	0.814	0.815	0.811
5	1.037	0.897	0.880	0.848	0.840	0.834	0.835	0.827	0.826	0.829	0.828	0.831	0.830	0.830	0.830	0.818	0.815	0.817	0.812
4	1.027	0.899	0.876	0.847	0.837	0.833	0.830	0.824	0.824	0.826	0.826	0.830	0.829	0.828	0.828	0.818	0.815	0.816	0.811
3	1.030	0.892	0.871	0.843	0.835	0.830	0.826	0.823	0.822	0.824	0.828	0.829	0.830	0.829	0.830	0.821	0.816	0.818	0.814
2	1.037	0.895	0.869	0.846	0.839	0.837	0.836	0.830	0.827	0.829	0.829	0.831	0.832	0.831	0.833	0.822	0.819	0.821	0.816
1	1.019	0.887	0.868	0.844	0.843	0.835	0.839	0.830	0.828	0.830	0.830	0.832	0.832	0.831	0.831	0.822	0.820	0.821	0.817

Coefficient-history pairs

Partial | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.099	0.923	0.891	0.868	0.860	0.852	0.846	0.837	0.834	0.833	0.835	0.838	0.839	0.836	0.836	0.823	0.820	0.821	0.815
7	1.082	0.922	0.886	0.863	0.853	0.847	0.843	0.835	0.831	0.832	0.834	0.838	0.838	0.836	0.834	0.823	0.820	0.820	0.814
6	1.077	0.916	0.883	0.858	0.848	0.842	0.839	0.833	0.830	0.831	0.832	0.835	0.836	0.835	0.834	0.822	0.818	0.819	0.813
5	1.065	0.911	0.884	0.854	0.847	0.839	0.836	0.830	0.829	0.829	0.831	0.835	0.835	0.835	0.834	0.822	0.819	0.819	0.813
4	1.054	0.903	0.876	0.849	0.842	0.836	0.834	0.828	0.828	0.830	0.830	0.835	0.835	0.834	0.833	0.822	0.819	0.821	0.815
3	1.049	0.909	0.878	0.847	0.845	0.839	0.839	0.831	0.829	0.829	0.830	0.833	0.834	0.834	0.834	0.824	0.820	0.822	0.816
2	1.043	0.907	0.881	0.854	0.850	0.844	0.843	0.831	0.831	0.832	0.831	0.834	0.834	0.833	0.833	0.825	0.823	0.823	0.818
1	1.015	0.887	0.867	0.844	0.842	0.834	0.838	0.829	0.828	0.829	0.829	0.831	0.831	0.831	0.830	0.822	0.820	0.821	0.817

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.808
7	0.808
6	0.807
5	0.809
4	0.811
3	0.811
2	0.813
1	0.812

Coefficient-history pairs

Mean target MSE (squared log units)

K=0 own-history baseline: one reference per L, shared across modes/rules

Own control